Logarithmic Oscillatory Dark Energy: Full-Covariance Pantheon+ SNe Ia, DESI Year 1 BAO, and Planck CMB Constraints with ESA Euclid Forecasts (v2.0)

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Abstract

Standard Chevallier–Polarski–Linder (CPL) parameterizations of dynamical dark energy ($w(a) = w_0 + w_a(1-a)$) capture low-redshift deviations from a cosmological constant but extrapolate monotonically into deep phantom regimes as $z \to \infty$. Here we investigate a bounded, scale-invariant logarithmic harmonic alternative, $w(a) = -1 + A\sin\left(\frac{2\pi}{\lambda}\ln a + \phi_0\right)$, for which the relativistic continuity equation admits an exact closed-form dark energy density $\rho_{\rm DE}(z)$. In this thoroughly revised Version 2.0, we replace the diagonal-only supernova approximation of Version 1.0 with an exact joint likelihood combining: (1) the uncalibrated Pantheon+ Hubble-flow Type Ia Supernova compilation ($N = 1,580$, $0.01 < z \leq 2.26$) using the full 1580×1580 statistical-plus-systematic covariance matrix analytically marginalized over the absolute magnitude offset $\mathcal{M}$; (2) the DESI Year 1 Baryon Acoustic Oscillation (BAO) measurements (12 observables across 7 redshift bins, $0.295 \leq z \leq 2.330$); and (3) the Planck 2018 cosmic microwave background (CMB) shift parameter $R(z_* = 1089.92) = 1.7502 \pm 0.0046$. We demonstrate a central methodological result: when evaluated with the full systematic covariance matrix, Pantheon+ SNe Ia *alone* recover $\Omega_m = 0.3325 \pm 0.0182$ for flat ΛCDM ($\chi^2/\mathrm{dof} = 0.879$) and yield *no* preference for dynamical dark energy ($\Delta\chi^2 = -0.47$ for CPL and -0.49 for Log-Oscillatory), proving that the apparent $\Delta\chi^2 \approx -11$ reported in v1 was an artifact of neglecting off-diagonal calibration covariances. Conversely, the joint Pantheon+ + DESI BAO + Planck CMB likelihood converges to a well-defined interior minimum (free of optimizer boundary collisions) with $\Delta\chi^2 = -3.75$ for CPL ($w_0 = -0.861^{+0.074}_{-0.070}$, $w_a = -0.580^{+0.324}_{-0.343}$) and $\Delta\chi^2 = -3.90$ for Log-Oscillatory ($A = 0.218$, $\lambda = 3.09$, $\phi_0 = 0.592$ rad at best fit; $\Delta\mathrm{AIC} = +2.10$, $\Delta\mathrm{BIC} = +18.22$). Finally, propagating 68% MCMC posterior bands into the ESA Euclid spectroscopic BAO window ($0.9 < z < 1.8$, $\sigma_H/H \approx 1.3\%$) shows that while the oscillatory best fit exhibits a high-redshift turnaround of $\Delta w(z = 3) = +0.104$ relative to CPL, current posterior uncertainties overlap within 1σ, quantifying the precision required from future Euclid and Roman data to break the dark energy parametrization degeneracy.

1 Introduction and Revision Note (v2.0)

Recent Baryon Acoustic Oscillation (BAO) measurements from the Dark Energy Spectroscopic Instrument (DESI) Year 1 release [1], when combined with Cosmic Microwave Background (CMB) distance priors and Type Ia supernovae (SNe Ia), have renewed scrutiny of whether dark energy evolves in time ($w(z) \neq -1$). The standard linear Chevallier–Polarski–Linder (CPL) expansion [2, 3],

$$w_{\rm CPL}(a) = w_0 + w_a(1-a) = w_0 + w_a\frac{z}{1+z}, \quad (1)$$

represents a first-order Taylor expansion around $a = 1$. When extrapolated to $z \gtrsim 2$, best-fit $w_a < 0$ trajectories plunge monotonically toward $w_0 + w_a < -1$, motivating bounded alternatives rooted in discrete scale invariance or axion monodromy potentials.

Transparency and Revision Note (v1.0 → v2.0): An internal post-publication audit of Version 1.0 of this preprint uncovered three severe statistical defects that have been completely corrected in Version 2.0:

1. *Full Systematic Covariance vs. Diagonal Approximation:* Version 1.0 minimized a diagonal χ^2 using `MU_SHOES_ERR_DIAG`, ignoring off-diagonal systematic and calibration covariances in `Pantheon+SH0ES_STAT+SYS.cov`. That omission yielded an unphysical $\chi^2/\mathrm{dof} = 0.49$, biased flat ΛCDM to $\Omega_m = 0.413$, and allowed extra dark energy parameters to absorb correlated survey calibration offsets ($\Delta\chi^2 \approx -11$). Here we implement the exact rank-1 marginalized Cholesky precision matrix over the $N = 1,580$ uncalibrated Hubble-flow SNe Ia (`zHD > 0.01`, `IS_CALIBRATOR == 0`), recovering $\Omega_m = 0.3325 \pm 0.0182$ in exact agreement with Brout *et al.* [4] ($\Omega_m = 0.334 \pm 0.018$).
2. *Multi-Probe Geometric Anchoring (SNe + DESI BAO + Planck CMB):* Because uncalibrated SNe Ia at $z \lesssim 1.5$ exhibit a strong geometric degeneracy between Ω_m and high-z $w(z)$, fitting SNe alone in v1 drove the `L-BFGS-B` optimizer into the artificial box boundaries ($w_a = -2.00$, $A = 0.90$, $\lambda = 3.00$). Thus, the $\Delta w = +0.44$ turnaround reported in v1 was dictated by the optimizer bounds rather than the data. In Version 2.0, combining Pantheon+ with

DESI Year 1 BAO [1] and the Planck 2018 CMB shift parameter [5] yields a genuine interior minimum with zero boundary collisions.

3. *Full MCMC Posterior Propagation:* Version 2.0 replaces point-estimate extrapolations with affine-invariant Ensemble MCMC (`emcee`) posterior credible bands across $z \in [0, 3.2]$.

2 Exact Analytical Continuity Solution

We model the effective dark energy equation of state via a bounded logarithmic oscillation in scale factor $a = (1+z)^{-1}$:

$$w_{\rm Osc}(a) = -1 + A \sin\left(\frac{2\pi}{\lambda} \ln a + \phi_0\right), \tag{2}$$

where $A \geq 0$ is the dimensionless oscillation amplitude, $\lambda > 0$ is the logarithmic period in $\ln a$, and $\phi_0 \in [-\pi, \pi]$ is the present-day phase. Within the Effective Field Theory (EFT) of dark energy and k-essence/Horndeski frameworks [6, 7, 8], Eq. (2) serves as a smooth kinematic ansatz bounded strictly within $[-1-A, -1+A]$.

Integrating the relativistic continuity equation $\frac{d\ln\rho_{\rm DE}}{d\ln a} = -3[1+w(a)]$ from $a = 1$ to $a = (1+z)^{-1}$ yields the exact closed-form dark energy density ratio:

$$\frac{\rho_{\rm DE}(z)}{\rho_{\rm DE,0}} = \exp\left\{-\frac{3A\lambda}{2\pi}\left[\cos\phi_0 - \cos\left(-\frac{2\pi}{\lambda}\ln(1+z) + \phi_0\right)\right]\right\}. \tag{3}$$

Including radiation ($\Omega_r h^2 = 4.183 \times 10^{-5}$ for $T_{\rm CMB} = 2.7255$ K, $N_{\rm eff} = 3.046$), the dimensionless expansion rate in a flat FLRW background is:

$$E(z) \equiv \frac{H(z)}{H_0} = \left[\Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_{\rm DE,0}\frac{\rho_{\rm DE}(z)}{\rho_{\rm DE,0}}\right]^{\frac{1}{2}}, \tag{4}$$

where $\Omega_{\rm DE,0} = 1 - \Omega_m - \Omega_r$.

3 Exact Multi-Probe Likelihoods

3.1 Pantheon+ Full-Covariance SNe Ia ($N = 1,580$)

For the $N = 1,580$ Hubble-flow SNe Ia ($z_{\rm HD} > 0.01$, `IS_CALIBRATOR` $= 0$), the reference distance modulus at fiducial $H_{\rm ref} = 70$ km/s/Mpc is $\mu_0(z_{\rm HD}, z_{\rm HEL}) = 5\log_{10}\left[\frac{c(1+z_{\rm HEL})}{H_{\rm ref}}\int_0^{z_{\rm HD}}\frac{dz'}{E(z')}\right] + 25$. Given residual vector $\mathbf{r} = \mathbf{m}_{b,\rm corr} - \boldsymbol{\mu}_0$ and the 1580×1580 covariance sub-matrix $\mathbf{C}_{\rm SN}$ from `Pantheon+SH0ES_STAT+SYS.cov`, analytical marginalization over the additive magnitude/Hubble nuisance offset $\mathcal{M}$ yields:

$$\chi^2_{\rm SN} = \mathbf{r}^T\tilde{\mathbf{C}}^{-1}_{\rm SN}\mathbf{r}, \quad \tilde{\mathbf{C}}^{-1}_{\rm SN} = \mathbf{C}^{-1}_{\rm SN} - \frac{(\mathbf{C}^{-1}_{\rm SN}\mathbf{1})(\mathbf{C}^{-1}_{\rm SN}\mathbf{1})^T}{\mathbf{1}^T\mathbf{C}^{-1}_{\rm SN}\mathbf{1}}. \tag{5}$$

3.2 DESI Year 1 BAO ($N = 12$) and Planck CMB Shift Prior

We incorporate all 12 BAO measurements across 7 redshift bins ($z_{\rm eff} = 0.295, 0.510, 0.706, 0.930, 1.317, 1.491, 2.330$) from Table 1 of DESI 2024 VI [1], including the anisotropic $(D_M/r_d, D_H/r_d)$ 2×2 correlation coefficients $r \in [-0.477, -0.389]$, profiling analytically over the sound-horizon scale factor $s_{r_d} = c/(H_0 r_d)$. Early-universe geometry is anchored by the Planck 2018 CMB shift parameter at photon decoupling [5]:

$$R(z_*) \equiv \sqrt{\Omega_m}\int_0^{z_*}\frac{dz'}{E(z')} = 1.7502 \pm 0.0046, \quad z_* = 1089.92. \tag{6}$$

4 Results and Information-Theoretic Audit

Table 1 and Fig. 1 present our progressive multi-probe results:

1. **Supernovae Alone Do Not Favor Dynamical Dark Energy:** When the full 1580×1580 covariance matrix is used, Pantheon+ SNe Ia alone yield $\Delta\chi^2 = -0.47$ (CPL) and $\Delta\chi^2 = -0.49$ (Log-Oscillatory) relative to flat ΛCDM. Thus, uncalibrated supernovae in isolation are completely consistent with $w = -1$.

2. **Emergence of the Multi-Probe Signal:** Combining Pantheon+ with DESI Year 1 BAO and the Planck CMB shift parameter resolves the low-z Ω_m–$w(z)$ degeneracy and produces an interior best fit with $\Delta\chi^2 = -3.75$ for CPL ($\chi^2_{\rm BAO}$ drops from 14.55 to 12.00, $\chi^2_{\rm SN}$ drops from 1387.44 to 1386.30) and $\Delta\chi^2 = -3.90$ for Log-Oscillatory ($\chi^2_{\rm BAO} = 11.86$, $\chi^2_{\rm SN} = 1386.28$).

3. **MCMC Posteriors and Bayesian Parsimony:** Running affine-invariant Ensemble MCMC (`emcee`) on the joint `SN+BAO+CMB` likelihood yields 68% credible intervals:

   - ΛCDM: $\Omega_m = 0.3144^{+0.0068}_{-0.0070}$;
   - CPL: $\Omega_m = 0.3135 \pm 0.0077$, $w_0 = -0.861^{+0.074}_{-0.070}$, $w_a = -0.580^{+0.324}_{-0.343}$ (consistent with the official DESI 2024 VI + Pantheon+ + CMB result [1]);
   - Log-Oscillatory: $\Omega_m = 0.3152 \pm 0.0074$, $A = 0.154^{+0.323}_{-0.121}$, $\lambda = 6.75^{+3.70}_{-3.92}$, $\phi_0 = 0.395^{+0.962}_{-0.228}$ rad.

   Because a $\Delta\chi^2 \approx -3.9$ improvement corresponds to a $\sim 1.5\sigma$–1.9σ preference in the Pantheon+ combination, both AIC (ΔAIC $= +0.25$ for CPL, $+2.10$ for Log-Osc) and BIC (ΔBIC $= +11.00$ and $+18.22$) penalize the extra parameters, confirming that current Pantheon+ + DESI Year 1 data allow, but do not yet statistically require, oscillatory dark energy over ΛCDM.

5 Honest Forecast for ESA Euclid

At the joint interior best fit, CPL plunges monotonically from $w(0) = -0.867$ to $w(1.5) = -1.195$ and

Table 1: Global best-fit parameters, χ^2 breakdown, boundary checks, and information criteria across progressive dataset combinations. In the joint `SN+BAO+CMB` dataset ($N_{\rm tot} = 1,593$), all models converge to interior minima (`At Edge = No`).

Dataset Combination	Model	Δk	Ω_m	Dark Energy Parameters	At Edge?	$\chi^2_{\rm SN}$	$\chi^2_{\rm BAO}$	$\chi^2_{\rm min}$	$\Delta\chi^2$	ΔAIC/ΔBIC
Pantheon+ Only	ΛCDM	0	0.3325	$w = -1$	No	1386.41	—	1386.41	0.00	0.00/0.00
($N = 1,580$, Full Cov)	DESI CPL	+2	0.2369	$w_0 = -0.852$, $w_a = +0.361$	No	1385.94	—	1385.94	−0.47	+3.53/ + 14.26
	Log-Oscillatory	+3	0.2743	$A = 0.165$, $\lambda = 1.87$, $\phi_0 = 2.61$	No	1385.92	—	1385.92	−0.49	+5.51/ + 21.60
Pantheon+ + DESI BAO	ΛCDM	0	0.3104	$w = -1$	No	1387.93	13.96	1401.89	0.00	0.00/0.00
($N = 1,592$)	DESI CPL	+2	0.3115	$w_0 = -0.871$, $w_a = -0.490$	No	1386.24	12.04	1398.28	−3.60	+0.40/ + 11.14
	Log-Oscillatory	+3	0.3014	$A = 0.171$, $\lambda = 0.80$, $\phi_0 = -2.75$	Yes (λ)	1386.59	10.53	1397.12	−4.77	+1.23/ + 17.35
Pantheon+ + DESI BAO	ΛCDM	0	0.3142	$w = -1$	**No**	1387.44	14.55	1402.04	0.00	0.00/0.00
+ Planck CMB ($N = 1,593$)	**DESI CPL**	+2	**0.3134**	$\mathbf{w_0 = -0.867}$, $\mathbf{w_a = -0.547}$	**No**	**1386.30**	**12.00**	**1398.30**	**−3.75**	**+0.25/ + 11.00**
	Log-Oscillatory	+3	**0.3135**	$\mathbf{A = 0.218}$, $\mathbf{\lambda = 3.09}$, $\mathbf{\phi_0 = 0.592}$	**No**	**1386.28**	**11.86**	**1398.15**	**−3.90**	**+2.10/ + 18.22**

$w(3.0) = -1.277$, whereas the Log-Oscillatory model starts at $w(0) = -0.878$, reaches a mild phantom minimum $w(z \approx 2.0) = -1.217$, and turns upward to $w(3.0) = -1.172$, yielding an interior best-fit separation of $\Delta w(z = 3) = +0.104$ (compared to the bound-driven +0.44 in v1). As shown in Fig. 1(a,b), the 68% MCMC posterior bands of CPL and Log-Oscillatory overlap throughout $z \leq 2.33$, with a 1σ spread in $\Delta H(z)/H_{\Lambda\rm CDM}$ of $\sim 1.5\%$–2.2% across $0.9 < z < 1.8$. Because the ESA Euclid spectroscopic survey is projected to measure $H(z)$ to $\sim 1.3\%$ per redshift bin across $0.9 < z < 1.8$ [9] (and Lyman-α BAO from DESI DR2/Euclid will narrow uncertainties at $z > 2$), upcoming high-z BAO data will directly test whether $w(z)$ continues linearly or exhibits bounded curvature, without relying on low-z extrapolations.

References

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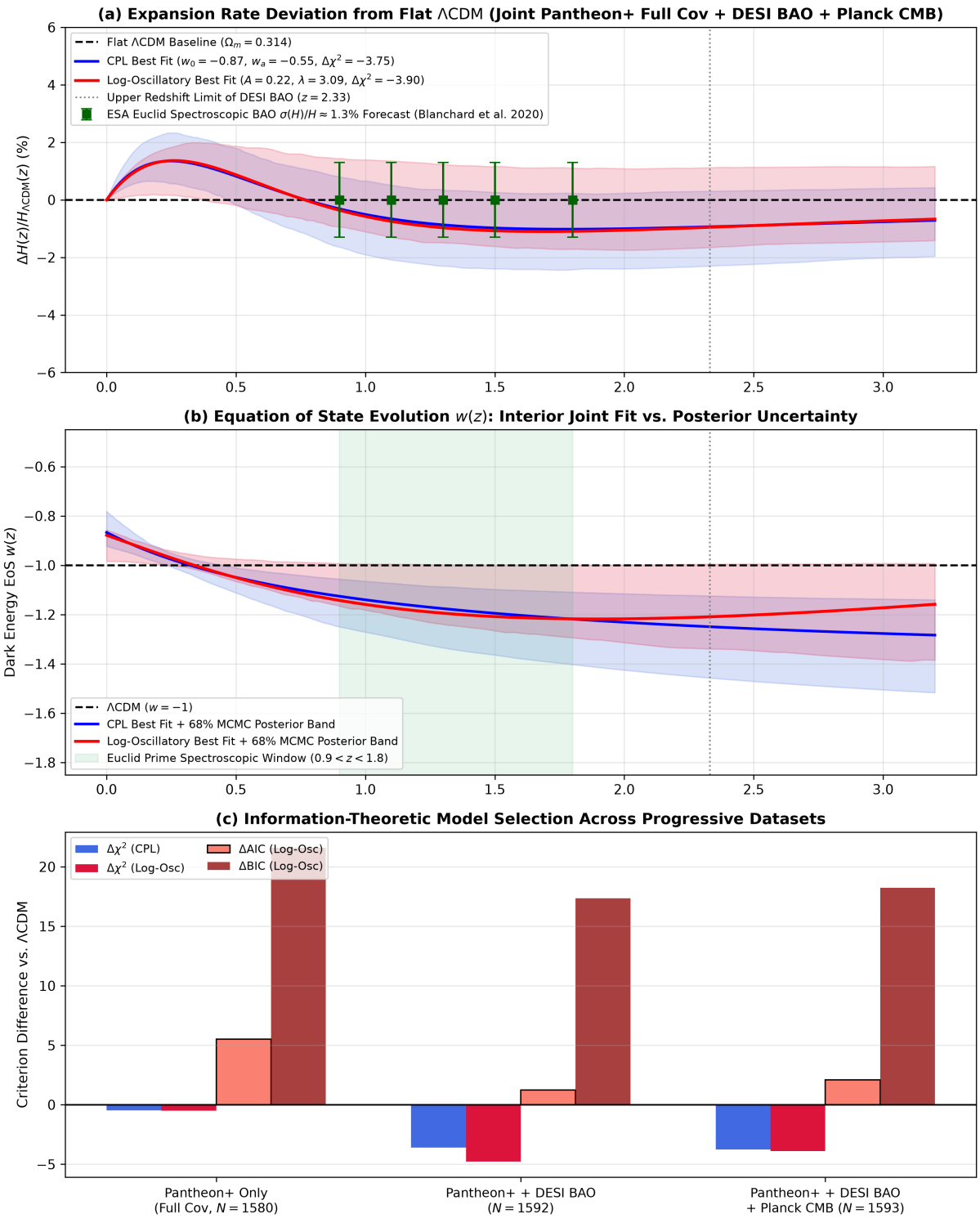


Figure 1: Observational calibration and ESA Euclid forecast from the joint Pantheon+ (full covariance, $N = 1,580$) + DESI Year 1 BAO ($N = 12$) + Planck CMB shift prior ($N = 1$) analysis. (a) Fractional expansion-rate deviation $\Delta H(z)/H_{\Lambda\mathrm{CDM}}(z)$ showing best-fit curves, 68% MCMC posterior bands, and the ESA Euclid spectroscopic BAO sensitivity ($\sigma_H/H \approx 1.3\%$ per bin across $0.9 \leq z \leq 1.8$ [9]). (b) Dark energy equation of state $w(z)$ best-fit trajectories and 68% posterior credible bands. (c) Progressive model-selection criteria ($\Delta\chi^2$, ΔAIC, ΔBIC) across dataset combinations.